Same Evidence, Different Skill: State-Regeneration Instability in Self-Evolving Agents

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Paper under double-blind review

Abstract

Self-evolving agents convert execution feedback into persistent skills, but most studies emphasize which trajectories reach the updater rather than whether the updater reliably materializes a useful persistent state. We study a narrower failure mode: byte-identical learner-visible evidence can fail to regenerate the same behaviorally useful skill. A controlled DeepSeek-V4-Pro study of 48 matched evidence pairs, 96 learned states, and 1,728 held-out evaluations first shows that replacing winner evidence with a mixed rejected-witness view has a small heterogeneous mean effect of 0.023, with a frozen confidence interval crossing zero. We then examine one outcome-selected development stream. A historically strong First-Fail state remains directionally useful across frozen-state actor remeasurements, whereas two fresh updater states produced from reconstructed byte-identical First-Fail evidence do not reproduce the historical advantage. These observations rule out evidence disappearance and weaken a pure one-rollout actor-luck explanation, while remaining consistent with a persistent-state generation bottleneck; they do not identify a population variance component. We therefore treat state generation as an experimental variable and define a prospective typed compiler intervention that maps the same trajectory+score package to a finite repair representation and canonical skill state. A fresh Evidence$\times$Generator bridge independently tests the matched-evidence generator effect, trajectory-conditioned content against score-only and scope-matched generic controls, and any secondary First-Fail evidence effect. The present paper establishes local state-regeneration instability and a falsifiable generator hypothesis; it does not yet claim that the compiler improves utility or that rejected failures are uniquely beneficial.

1 Introduction

Self-evolving agents improve without updating model weights by turning execution experience into persistent procedural state. A spreadsheet agent, for example, may execute several attempts of the same task, retain successful or failed trajectories, and ask an LLM updater to revise a reusable skill document for future tasks. This paradigm is attractive because it moves adaptation into an inspectable persistent artifact rather than another round of model training. Recent systems already show that failed trajectories can matter, that success/failure contrasts can reveal useful corrections, and that accumulated execution knowledge can support later skill evolution (Liu et al., 2026b; Chen et al., 2026; Liu et al., 2026a; Tang et al., 2026).

A common assumption nevertheless remains implicit: once the updater is given the "right" evidence, writing the persistent state is mostly a downstream implementation step. Under this view, the scientific question is which trajectory or feedback view should reach the learner. Our experiments challenge that reduction. In a controlled study with the same actor, verifier, updater implementation, initial skill, and held-out evaluator, replacing winner evidence with a rejected-witness view does not yield a stable population-level advantage. More importantly, in one outcome-selected development stream, a persistent state generated from First-Fail evidence remains useful when its bytes are frozen, yet fresh updater runs from reconstructed byte-identical evidence fail to recreate the same advantage. The useful evidence did not disappear; the useful state was not reliably regenerated.

This distinction matters because persistent skill text is itself an intervention on future behavior. Two updater calls can receive the same evidence and still synthesize states that differ in scope, redundancy, ordering, specificity, or procedural commitments. If downstream utility is sensitive to those differences, then "which evidence was shown?" and "how was that evidence materialized into state?" are separate experimental variables. A self-evolution pipeline can therefore fail even when evidence selection is held fixed.

We organize the system as

$$\text{search trajectories} \rightarrow \underbrace{E}_{\text{evidence selection}} \rightarrow \underbrace{G}_{\text{state generation}} \rightarrow \underbrace{S}_{\text{persistent state}} \rightarrow \underbrace{Y}_{\text{future utility}} . \tag{1}$$

This decomposition is not itself a novelty claim. Closely related systems already separate traces, diagnosis, persistent knowledge, skill revision, and execution (Liu et al., 2026a; Chen et al., 2026; Tang et al., 2026). Its role here is operational: it lets us hold learner-visible evidence fixed while identifying the realized persistent state and changing or freezing the state-generation mechanism. The residual scientific object is the *state-regeneration outcome*: what persistent artifact and downstream behavior are obtained when the same evidence is passed through the updater again.

Our evidence follows that intervention logic rather than a chronology of increasingly complex methods. First, a 48-pair controlled study shows that rejected-witness evidence is not uniformly superior to winner evidence. Second, a single-stream witness-selection intervention fails to recover a simple "more diagnostic failure is better" law. Third, the historically strong First-Fail state remains directionally useful under frozen-state actor remeasurements, weakening a pure one-rollout downstream-noise explanation. Fourth, two fresh updater realizations from byte-identical First-Fail evidence fail to reproduce the historical advantage. This does not estimate a population variance component: the development case was selected after outcomes, only two fresh updater states exist, and their original measurements used fresh actor realizations. The completed evidence is therefore narrower: *local state-regeneration instability consistent with a state-generation bottleneck*.

That observation motivates a controlled intervention on G. We use a small typed state compiler that maps learner-visible trajectory+score evidence into a finite diagnosis/repair representation and canonical skill state. The point is not that structured diagnosis or deterministic text editing is novel—SkillRevise already performs execution-grounded diagnosis and repair, SkillCAT uses same-task success/failure contrasts, and WikiSkill separates raw experience from persistent knowledge (Liu et al., 2026a; Chen et al., 2026; Tang et al., 2026). Nor do three spreadsheet repair primitives constitute a broad standalone algorithm. The compiler is first an *experimental intervention*: it deliberately removes free-form state-synthesis degrees of freedom while preserving the learner-visible evidence package.

We preregister a fresh development bridge that independently tests four questions rather than forcing them into one pass/fail story: (i) a matched-evidence generator-method effect, (ii) whether trajectory-conditioned compilation beats score-only and state-scope-matched trajectory-blind generic controls, (iii) any secondary First-Fail evidence-package effect, and (iv) an Evidence$\times$Generator interaction. SCREEN and VALIDATION use different streams and disjoint held-out panels. A generator effect may proceed to validation even if First-Fail evidence is not better than winner evidence; failure-evidence and interaction claims are secondary. A small repeated-synthesis diagnostic on validation is separately reserved for the stronger state-regeneration/reproducibility interpretation. None of these prospective stages is reported as a positive scientific result here.

Contributions. The current paper makes two completed contributions and defines two prospective falsification objects.

- **A matched-evidence state-regeneration audit.** We identify persistent state bytes as an explicit experimental object and show, in a selected controlled case, that a historically useful state remains behaviorally active when frozen while byte-identical evidence does not reliably regenerate the same advantage through the native updater.
- **A bounded localization result.** Together with a 48-pair heterogeneous evidence-source study and a failed diagnostic-witness selector, the audit rules out evidence disappearance and weakens a pure one-rollout actor-luck account, while leaving regression-to-the-mean, selected-case bias, evaluator alignment, and population-level variance unresolved.

- **A controlled generator intervention.** We define a typed canonical compiler with information parity, content-addressed state identity, a score-only generic maximum, and a trajectory-blind state-scope-matched generic control. Its purpose is to isolate a complete state-generation-method effect, not to claim a new universal diagnosis language.
- **A hierarchical prospective test.** A fresh Evidence×Generator bridge validates the generator question independently of First-Fail superiority or interaction, with explicit STOP rules that prohibit E3, another backbone, or another benchmark from rescuing a failed fresh generator result.

2 Problem Setup and Closest Work

2.1 Persistent skill self-evolution

For an update stream u, the agent executes eight tasks. Task i produces a frozen search pool $P_i = \{\tau_{ij}\}_{j=1}^{K}$ with binary verifier scores $r_{ij} \in \{0, 1\}$. A serving rule chooses an acting winner for the user-facing task, while an evidence projection E determines which trajectory+score package is exposed to the persistent learner. The state generator G maps the eight learner-visible evidence units and initial skill S_0 to a new persistent skill,

$$S = G(E(P_1), \ldots, E(P_8); S_0). \tag{2}$$

A frozen downstream actor then executes held-out task x conditioned on S, producing utility $Y(x; S) \in \{0, 1\}$ under the deterministic spreadsheet verifier.

This decomposition creates three distinct estimands. The *evidence-package effect* compares different E while holding the generator fixed. The *state-generation-method effect* compares different G on the same selected evidence. The *state effect* holds the state bytes fixed and remeasures downstream behavior. These are not interchangeable. In particular, a useful frozen state does not imply that the updater can reliably regenerate it, and an evidence source that sometimes yields a useful state does not imply that the evidence source is generally superior.

2.2 Controlled spreadsheet substrate

Our mechanism studies use a locally generated SpreadsheetBench-compatible suite with six predeclared failure families: input/output contract, target sheet/range, schema/key alignment, aggregation/join, formula/materialization, and multi-step pipeline. Each task has deterministic initialized and golden workbooks and a deterministic verifier. This suite is used for causal control, not as a public benchmark. SpreadsheetBench itself contains real-world spreadsheet manipulation tasks and OJ-style evaluation (Ma et al., 2024); external benchmark claims are deliberately deferred until the internal mechanism survives.

The primary actor and updater model in the current mechanism line are DeepSeek-V4-Pro under a frozen Ark route. Actor decoding uses temperature zero, thinking disabled, and no provider retry. The first-party persistent updater is the MindMemOS SkillEvolver path used throughout the controlled study. Exact model routes, skill hashes, trajectory hashes, budgets, and task IDs are bound in execution receipts rather than inferred from manuscript prose.

2.3 Closest work and residual scientific object

Failure-aware feedback and skill evolution. Rethinking Self-Evolving Agent Skills varies the feedback presented to a skill optimizer and shows that failure-containing feedback is often selected, but the ranking of Normal, Fail-only, and Success-only conditions depends on model and benchmark (Liu et al., 2026b). This directly rules out a novelty claim that failures can help. Our residual question is different: with evidence bytes fixed, can the state-writing step itself become an unstable bottleneck?

Contrastive trajectory diagnosis. SkillCAT samples multiple trajectories, contrasts same-task successes and failures, extracts candidate corrections, and validates patches before merging (Chen et al., 2026). It therefore already covers success/failure pairing and contrastive causal extraction. Our

First-Fail evidence arm is not presented as a new contrastive-learning idea. We instead use matched evidence arms to factor evidence selection from the generator that materializes persistent state.

Execution-grounded skill revision. SkillRevise diagnoses skill defects from execution traces, retrieves reusable repair principles, and applies execution-anchored revisions that are re-executed and selected by utility (Liu et al., 2026a). Consequently, "trajectory $\rightarrow$ structured diagnosis $\rightarrow$ skill edit" is not our contribution. Our typed compiler is scientifically useful only if it tests a different claim: whether replacing unconstrained state synthesis with canonical compilation changes state reproducibility or downstream utility under the same evidence.

Persistent knowledge accumulation. WikiSkill separates raw execution experience, an accumulated persistent wiki, and executable skills, and uses the knowledge store to guide future skill evolution (Tang et al., 2026). We therefore do not claim novelty for compiling experience into persistent knowledge. The missing object here is narrower: the realized persistent state as a causal intermediate whose variability can be studied independently of evidence selection.

Residual novelty boundary. The paper's novelty target is therefore not a new feedback source, diagnosis vocabulary, memory store, or validation-gated text editor. It is the controlled matched-evidence study of state regeneration itself: a frozen-state audit asks whether behavior follows persistent state identity rather than one actor draw, and the prospective bridge varies the complete state-generation method while holding learner-visible evidence fixed. The existing case is consistent with a state-generation bottleneck but does not identify it at population scale. If the fresh generator contrast fails, the method-positive story stops. If the compiler helps but First-Fail evidence does not, the failure-learning claim is dropped while the state-generation-method result may remain.

3 From Evidence Heterogeneity to a State-Regeneration Anomaly

We use a sequence of interventions that progressively hold more of the pipeline fixed. The goal is not to turn one selected positive case into a general claim; it is to determine which layer remains plausible after simpler explanations fail.

3.1 Evidence-source effects are small and heterogeneous at scale

Our closed DeepSeek V2 study contains 48 matched pairs, 96 learned states, and 1,728 held-out task evaluations. The two primary states differ in the evidence projection exposed to the same frozen updater. Across pairs, the mean MRW$-$WIN-C utility difference is $+0.0231$, the median is $+0.0139$, and the standard deviation is 0.0780. The frozen one-sided sign-flip test gives $p = 0.1719$, and the bootstrap confidence interval is $[-0.0185, 0.0660]$.

This study therefore does not establish rejected-witness superiority. Individual streams span positive, null, and negative effects, including values from -0.125 to $+0.181$. The appropriate inference is heterogeneity, not a failed attempt that can be repaired by silently adding samples or widening a threshold.

3.2 Selecting a more "diagnostic" failure does not solve the problem

Post-hoc development analysis suggested that progress-related features correlated with pairwise effects, motivating a single-stream witness-selector pilot on `e1-tsr-00`. This stream was chosen after outcome inspection and is therefore mechanism development only. We froze four evidence arms: WIN-C, First-Fail, Progress-Fail, and Progress-Contrast. On 18 held-out tasks they score 13/18, 17/18, 14/18, and 14/18 respectively.

Although the historical First-Fail state is strong, the preregistered purpose of this pilot was to test whether a progress-based selector provides a stable repair rule. It does not. The frozen verdict is `S1_SIGNAL_FAIL_STOP_NO_S2`. We therefore reject the simple story that "choose a later or more diagnostic failure" explains the earlier heterogeneity.

Table 1: Evidence ladder. The independent unit and scientific interpretation change across stages; no row is promoted beyond its design.

Stage	Held fixed	Observation	Supported interpretation
V2 48-pair study	actor/updater configuration, task design	mean MRW−WIN-C = +0.0231, CI crosses 0	evidence-source effect is heterogeneous
S1 selector pilot	selected stream, updater path	13/18, 17/18, 14/18, 14/18	simple progress selector fails
Frozen-state remeasure	persistent state bytes	FF: 15/18, 16/18; WIN-C: 14/18, 12/18	strong state has reproducible local behavioral value
Exact-evidence replay	learner-visible evidence bytes	FF: 15/18, 11/18; WIN-C: 15/18, 12/18	same evidence does not reliably regenerate strong state

3.3 The strong persistent state remains useful when frozen

The next intervention holds the persistent state bytes fixed and re-runs only downstream evaluation. The historical strong First-Fail state is compared with the corresponding WIN-C state in two fresh complete actor replicates on the same 18-task development panel. Replicate 1 gives 15/18 versus 14/18; replicate 2 gives 16/18 versus 12/18.

These remeasurements do not make the selected stream representative, and they do not prove that First-Fail evidence caused the state. They do weaken a simpler explanation: the original 17/18 First-Fail result was not merely one lucky downstream actor realization attached to an otherwise inert state. The frozen artifact itself carries behaviorally relevant information.

3.4 Byte-identical evidence fails to regenerate the same advantage

We then reconstruct the exact S1 First-Fail evidence rendered to the updater. All 16 rendered packet hashes match the historical inputs. From this byte-identical evidence we create two new free-form updater realizations, then evaluate each fresh state against a fresh WIN-C realization on the same 18 tasks. Replicate 1 is 15/18 versus 15/18; replicate 2 is 11/18 versus 12/18. The mean fresh contrast is -0.0278.

The contrast with frozen-state remeasurement is the key anomaly. Holding the strong state fixed preserves a directional advantage; holding the evidence fixed but regenerating the state does not. Because the fresh updater states and their downstream evaluations both vary, this is not a variance-component decomposition and does not by itself isolate the updater from actor noise. It nevertheless rules out evidence disappearance as a sufficient explanation of the strong historical state and motivates a direct frozen-state regeneration audit.

3.5 What the completed evidence supports

The current evidence supports the following bounded statement:

> In one controlled development case, identical trajectory evidence did not reliably regenerate the historical behaviorally useful persistent skill, while the previously generated strong state retained downstream utility when frozen. The observation is consistent with persistent-state generation as a candidate bottleneck and cannot be explained by evidence disappearance alone.

It does *not* establish that rejected failures are generally better than winners, that free-form updater variance dominates actor variance in the population, or that deterministic compilation is already superior. These stronger claims require prospective interventions rather than retrospective interpretation of the selected case.

4 Typed State Compilation as a Controlled Generator Intervention

The state-regeneration result motivates an intervention on G, the state-generation interface. The compiler is intentionally modest: it is designed first to remove free-form synthesis degrees of freedom under matched evidence, not to claim a universal agent-repair algorithm. It receives the same learner-visible trajectory text and selected binary verifier score as the corresponding free-form updater arm, but replaces open-ended state writing with a finite diagnosis and canonical compilation path.

4.1 Design requirements

A valid generator intervention must satisfy four requirements.

R1: Information parity. The compiler may read only the selected trajectory text and score exposed to the corresponding free-form arm. It cannot condition on the experimental arm, projection name, stream/family label, task family, golden answer, held-out result, historical S1 outcome, or hidden provenance. SHA identifiers are retained only for integrity receipts.

R2: Explicit intermediate representation. The method exposes what it inferred before changing the skill. Otherwise a different free-form prompt simply moves the uncontrolled synthesis step. We represent each evidence unit by

$$d = (\text{failure stage}, \text{failed invariants}, \text{observed evidence}, \text{required repairs}). \tag{3}$$

R3: Canonical state construction. Identical diagnosis bundles must produce byte-identical persistent skills. Duplicate evidence must not increase state length, primitive ordering is fixed, and redundant repair surfaces are removed deterministically.

R4: Generic and scope falsification. A positive trajectory-conditioned result must beat controls that receive no trajectory text but are allowed the same score pattern, and—for the stronger diagnosis claim—the same rendered repair-block count. This prevents a shorter or less interfering state from being mistaken for correct semantic diagnosis.

4.2 Typed diagnosis from learner-visible evidence

The current intervention uses three repair primitives:

$$\mathcal{R} = \{\texttt{VERIFY\_OUTPUT}, \texttt{COMPLETE\_WORKFLOW}, \texttt{RECOVER\_TOOL\_ERROR}\}. \tag{4}$$

For a selected failed trajectory, visible signals map to repairs as follows:

- no materialized write or output save → `COMPLETE_WORKFLOW`;
- output saved but no visible reload/verification → `VERIFY_OUTPUT`;
- visible tool error without visible clean recovery → `RECOVER_TOOL_ERROR`.

A selected successful trajectory is not automatically labelled as needing a generic repair. Multiple repairs may compose.

These rules are deliberately interpretable and narrow. Completion, verification, and recovery are also plausible generic spreadsheet hygiene and closely aligned with the deterministic evaluator. We therefore do not present this vocabulary as standalone algorithmic novelty. Its scientific role is to instantiate a controlled generator whose representation and rendering behavior are fully specified.

4.3 Canonical compiler

For diagnoses $D = \{d_i\}_{i=1}^{8}$ and base skill S_0, compilation computes

$$C(D, S_0) = S_0 \oplus \text{Canon}\left(\bigcup_i d_i.\texttt{required\_repairs}\right). \tag{5}$$

Canonicalization uses a fixed primitive order and deduplicates repeated requirements. `COMPLETE_WORKFLOW` already includes save, reload, and target verification, so a separately rendered `VERIFY_OUTPUT` block is omitted when both are requested. `RECOVER_TOOL_ERROR` composes after completion semantics. Every diagnosis bundle and compiled skill is content-addressed.

The zero-provider implementation is qualified for deterministic compilation, duplicate suppression, hidden-label exclusion, and the three diagnosis routes. For diagnoses corresponding to the manually frozen mechanism interventions, it reproduces the G1 verification, G2 completion, and G3 completion+recovery skill surfaces byte-for-byte. This is implementation qualification, not evidence that those states improve future utility.

4.4 Two trajectory-blind generic controls

Score-only generic maximum. `SCORE_ONLY_GENERIC_MAX` receives the same ordered eight binary scores but no trajectory text. If all scores are one, it returns S_0. If any score is zero, it emits the maximal nonredundant generic v1 bundle, `COMPLETE_WORKFLOW+RECOVER_TOOL_ERROR`. It tests whether simply knowing that failures occurred plus injecting the strongest generic checklist explains the effect.

State-scope-matched generic maximum. The maximal generic bundle can be longer or broader than a trajectory-conditioned compiler state. We therefore additionally freeze `SCOPE_MATCHED_GENERIC_MAX`. It receives the same eight scores and exactly one scalar from the trajectory-conditioned compiler: the number $k \in \{0, 1, 2\}$ of nonredundant repair blocks that canonical compilation would render. It receives neither trajectory text nor primitive identity. A frozen trajectory-independent priority rule returns S_0 for $k = 0$, a one-block generic completion state for $k = 1$, and completion+recovery for $k = 2$.

The scope scalar is intentionally a control variable, not evidence supplied to the proposed method. If FF4_COMP beats the score-only maximum but not the scope-matched control, the result supports state sparsity/canonicalization at most; it does not establish trajectory-conditioned semantic diagnosis.

4.5 What the generator comparison identifies

The free-form and compiler paths are complete state-generation methods. They differ in representation, canonicalization, state scope, redundancy, and use of a generative writer. Thus a positive COMP−FREE contrast is a *complete generator-method effect*, not an isolated effect of determinism. State length, delta tokens, rendered block count, and skill SHA are recorded as mechanism diagnostics rather than post-hoc matched away.

A separate repeated-synthesis probe is needed for the stronger regeneration/reproducibility interpretation. It repeats the same free-form evidence package exactly once on pre-frozen validation streams and distinguishes between-state disagreement from within-frozen-state actor disagreement. Failure of that probe removes the state-generation-variability interpretation even if the complete compiler method improves utility.

5 Evaluation Contract

The remaining experiment program is staged so that a negative result cannot be rescued by changing the scientific question, opening an untouched benchmark, or sampling additional updater states after outcomes.

5.1 M2: deterministic state-semantic sufficiency

Before testing an automatic generator intervention, M2 asks whether the manually identified repair surfaces can causally alter downstream behavior at all. Four frozen persistent states share the same base spreadsheet skill:

- G0: the exact initial skill;
- G1: G0 plus output save/reload/target-cell verification;

- G2: G0 plus a complete Inspect→Read→Compute→Write→Save→Reload/Verify loop;
- G3: G2 plus clean recovery from malformed or failed tool calls.

No updater is called. Each state is evaluated on the same 18 development held-out tasks with DeepSeek-V4-Pro, temperature zero, thinking disabled, $K = 1$, ten turns, and the same deterministic verifier. The primary screen selects the simplest G1<G2<G3 whose utility exceeds G0 by at least $1/18$. Historical S1 outcomes are excluded from the gate. If the primary passes, a pre-frozen 17-task sensitivity excluding the one task split across provider quota eras must remain strictly positive; this sensitivity may veto but never rescue.

M2 establishes at most *semantic sufficiency*. G1–G3 were manually constructed after the selected historical state was inspected, so they do not show that an automatic learner can discover the useful repair.

5.2 M3: SAME-EVIDENCE FROZEN-STATE REGENERATION AUDIT

The previous M3 plan remeasured G0 plus the M2-selected manual arm. We replace it with a higher-information actor-noise test that directly targets the unresolved exact-evidence anomaly and requires no updater call.

We freeze four already-existing persistent artifacts:

- historical First-Fail state, SHA `97e28b4862...185252`;
- fresh exact-evidence First-Fail realization 1, SHA `596bd30b49...fdf04f`;
- fresh exact-evidence First-Fail realization 2, SHA `fb5454a27f...36615e`;
- one common WIN-C state, SHA `6df40f6170...c00649`.

The historical, replicate-1, and replicate-2 WIN-C updater artifacts are byte-identical at this SHA, so a single frozen comparator removes the earlier fresh-comparator-state instability.

All four states are evaluated contemporaneously on the exact same 18-task development panel with the frozen actor/runtime/verifier configuration and hash-balanced state order. The new M3 measurement contains no state synthesis. Its purpose is not to estimate a population variance component; it asks whether the behavioral separation among already-generated same-evidence First-Fail states persists when the downstream actor is re-run against one common state reference.

The predeclared interpretation is asymmetric. If the fresh FF1–FF2 ordering materially persists under frozen-state remeasurement while both are compared with the same WIN-C artifact, a one-draw actor explanation becomes substantially less plausible. If their separation collapses or reverses, the manuscript must downgrade the state-regeneration-instability claim before M4; M2 cannot rescue it because M2 tests manually authored semantics, not state-generation realization.

5.3 M4: FRESH EVIDENCE×GENERATOR BRIDGE

The automatic bridge uses a newly generated development suite disjoint from the selected S1 panel and from the untouched E3 confirmation lane. It contains 120 formal task artifacts: 96 update tasks arranged as 12 eight-task streams, 12 SCREEN held-out tasks, and 12 different VALIDATION held-out tasks. SCREEN and VALIDATION each contain one stream per controlled failure family, and their held-out panels are disjoint.

For each stream we freeze the 2×2:

Evidence package	Native free-form updater	Typed compiler
Winner	W_FREE	W_COMP
First-Fail-4	FF4_FREE	FF4_COMP

Winner evidence. All eight update pools expose the served winner trajectory+score package.

First-Fail-4 evidence. A pool is eligible only if it contains both success and failure. A stream is bridge-eligible only with at least four mixed pools. Exactly four eligible pools are selected by a pre-frozen content hash; those expose the deterministic first failed nonwinner, while the other four expose the same winner package as the Winner arm. This equalizes the *replacement count*, not trajectory length, action count, failure severity, semantic density, or information amount. S_C is therefore an evidence-package-policy effect, not a purified "failure versus success" causal effect.

Information parity. Within an evidence package, FREE and COMP receive the same selected trajectory bytes and binary score. The compiler cannot read hidden arm, family, task, rollout, or held-out labels. Corresponding learner-input hashes must match before provider execution.

Free-form stochasticity. The updater and actor use temperature zero, thinking disabled, and no provider retry, but the provider does not expose a reliable generation seed for every call. The first free-form state is therefore a prospectively drawn deployed state, not a deterministic artifact. No extra updater draw may replace an unfavorable primary state. On VALIDATION only, one additional same-evidence FF4 free-form synthesis is pre-authorized as a regeneration diagnostic and is generated regardless of the first outcome.

5.4 HIERARCHICAL SCREEN AUTHORITY

For stream-level held-out utility J, define

$$G_F = J(\text{FF4\_COMP}) - J(\text{FF4\_FREE}), \tag{6}$$
$$S_C = J(\text{FF4\_COMP}) - J(\text{W\_COMP}), \tag{7}$$
$$I = [J(\text{FF4\_COMP}) - J(\text{FF4\_FREE})] - [J(\text{W\_COMP}) - J(\text{W\_FREE})]. \tag{8}$$

The hypotheses are adjudicated separately.

Generator gate. The generator question passes SCREEN if mean $G_F > 0$, at least 4/6 streams have $G_F > 0$, and no technical or evidence-parity violation occurs. This gate tests the complete generator method under matched FF4 evidence.

Trajectory-conditioned-content gate. Separately, FF4_COMP must exceed both `SCORE_ONLY_GENERIC_MAX` and `SCOPE_MATCHED_GENERIC_MAX` at the frozen aggregate gate before the manuscript may attribute benefit to trajectory-conditioned diagnosis. If a generic control matches the compiler, the stronger diagnostic interpretation stops.

Failure-evidence and interaction gates. The First-Fail-specific story is secondary. It requires mean $S_C > 0$, $S_C > 0$ on at least 4/6 streams, and the predeclared positive interaction condition. Failure of S_C or I drops the First-Fail-specific claim but does *not* veto validation of an already positive generator effect.

VALIDATION authority for the generator paper therefore depends on the matched-evidence generator question and its load-bearing generic controls, not on First-Fail superiority or interaction. This resolves the earlier conjunctive gate that incorrectly made a valid generator result contingent on a separate evidence-source hypothesis.

5.5 VALIDATION AND THE DIRECT REGENERATION PROBE

If the corrected generator/content SCREEN gates pass, the same frozen methods are evaluated on six different validation streams and the 12 disjoint VALIDATION held-out tasks. The primary generator result uses the first prospectively drawn FF4 free-form state exactly as in SCREEN.

For the stronger regeneration interpretation, each VALIDATION stream additionally receives exactly one second free-form state, FF4_FREE_B, synthesized from the byte-identical FF4 evidence package. It is not a retry and cannot replace FF4_FREE_A. On a pre-frozen six-task variance panel (one task per controlled family), A, B, and the deterministic compiled state each receive exactly one additional actor remeasurement. Thus the probe distinguishes disagreement between independently synthesized FREE states from disagreement caused by re-running the actor on a frozen state.

For stream s, let $D_U(s)$ denote the average task-level disagreement between the actor-averaged utilities of FREE-A and FREE-B, and let $D_A(s)$ denote average within-frozen-FREE-state actor disagreement. A local regeneration-instability result requires mean $D_U - D_A > 0$ and positive $D_U - D_A$ on at least 4/6 validation streams. This is a small prospective localization, not a full variance-component estimate. If it fails, the paper must drop the claim that state-generation variability itself is demonstrated even if the complete compiler method improves utility.

5.6 Decision table and STOP rule

The paper does not require failure evidence to be superior in order to support a generator-method contribution.

- If the fresh matched-evidence generator gate fails on SCREEN or fails to replicate on VALIDATION, stop the automatic state-generation-method paper. Do not open E3, another backbone, or another public benchmark as rescue.
- If the generator wins but is matched by the trajectory-blind scope-matched generic control, stop the trajectory-conditioned compiler paper; any residual result is generic state scope/canonicalization.
- If the generator and controls pass but S_C or I fails, drop the rejected-failure claim and retain only the state-generation-method paper.
- If the generator method passes but the direct regeneration probe fails, retain a complete generator-method result but drop state-generation-variability/reproducibility as the demonstrated mechanism.
- Only a frozen fresh result that survives the relevant controls may motivate a separate untouched E3 proposal. Nothing in M4 itself grants E3 authority.

5.7 Scientific unit

The independent scientific unit is the updated stream, not the 12 held-out task executions or provider calls nested within a stream. The controlled suite is a mechanism substrate rather than external-validity evidence. No outcome-conditioned additional stream, state synthesis, actor replicate, backbone, or benchmark is permitted to rescue a failed frozen gate.

6 Current Results and Open Claim Gates

At this manuscript checkpoint, the large-sample evidence-source study, S1 selector pilot, historical frozen-state stability study, and exact-evidence replay are complete and outcome-readable. M2 Recovery V3 is an active frozen measurement object whose incomplete outcomes remain sealed. The revised M3 regeneration audit and M4 automatic bridge are prospective. We report only completed evidence here.

6.1 Rejected-witness evidence is not a stable universal treatment

The 48-pair DeepSeek V2 study does not pass its frozen superiority criterion. The mean rejected-witness-minus-winner contrast is $+0.0231$, with bootstrap interval $[-0.0185, 0.0660]$ and one-sided sign-flip $p = 0.1719$. The stream distribution includes positive, null, and negative effects. This directly narrows the paper: the central contribution cannot be "failed trajectories are better learning data."

6.2 A diagnostic-witness selector fails its development gate

On the selected `e1-tsr-00` stream, First-Fail is 17/18 while WIN-C is 13/18, but progress-based failure and progress-contrast states are both 14/18. Because the prospective question was whether a more progress-diagnostic selector repairs the instability, the result is a failure of that hypothesis rather than evidence for First-Fail as a general rule. No S2 selector expansion is opened.

Evidence source alone is unresolved. 48-pair MRW$-$WIN-C is small and heterogeneous.

$\Downarrow$

Simple witness selection fails. Progress-based selectors do not recover the strong First-Fail case.

$\Downarrow$

Freeze the state. The historical strong state remains directionally useful across two fresh actor remeasurements.

$\Downarrow$

Freeze the evidence, regenerate the state. Two byte-identical-evidence updater realizations fail to recover the historical advantage.

$\Downarrow$

Open causal question. Is this persistent-state regeneration instability beyond downstream actor noise, and can a controlled generator method improve fresh-state utility?

Figure 1: Current evidence ladder. Each step removes a simpler explanation but the completed evidence remains a selected-case regeneration anomaly rather than a population variance decomposition.

6.3 State freezing and evidence replay point to a regeneration anomaly

The historical strong First-Fail state remains directionally stronger than its WIN-C comparator when both states are frozen: 15/18 versus 14/18 in one complete remeasurement and 16/18 versus 12/18 in a second. The same historical advantage does not survive state regeneration from byte-identical First-Fail evidence. Fresh updater realization 1 yields 15/18 versus a fresh WIN-C run of 15/18; realization 2 yields 11/18 versus 12/18.

Figure 1 summarizes what these observations do and do not remove. Byte-identical evidence rules out evidence disappearance. Repeated evaluation of the historical strong state makes a pure one-lucky-actor-roll explanation less plausible. But regression-to-the-mean, selected-case bias, evaluator alignment, and fresh-comparator/actor variability remain. The completed result is therefore *state-regeneration instability in one controlled development case*, not a measured state-generation variance component.

6.4 The strongest current claim

The strongest manuscript-level statement supported today is:

> In one controlled development case, reconstructed byte-identical trajectory evidence did not reliably regenerate the historical behaviorally useful skill through the native free-form updater, while the historical state itself remained directionally useful when frozen and re-evaluated. This observation is consistent with a persistent-state generation bottleneck but does not by itself isolate generator variance from downstream actor noise or selected-case effects.

The manuscript does not currently claim that the typed compiler improves downstream utility, that First-Fail-4 is superior to winner evidence, that state-generation variability dominates actor variability, or that the mechanism generalizes beyond the controlled development substrate.

6.5 What the revised next gates decide

M2 remains a semantic-sufficiency probe for the manually identified G1–G3 surfaces. The revised M3 then remeasures the existing historical First-Fail, fresh First-Fail-1, fresh First-Fail-2, and one byte-identical common WIN-C state without any new updater call; this directly attacks the actor-noise/comparator-instability objection in the completed evidence.

M4 no longer requires the First-Fail source effect and Evidence$\times$Generator interaction to pass before the generator question may validate. A fresh matched-evidence compiler advantage can proceed independently. The stronger trajectory-conditioned diagnosis claim additionally requires superiority over the score-only and state-scope-matched trajectory-blind generic controls. A small same-evidence repeated-synthesis probe on validation determines whether a state-regeneration/reproducibility interpretation survives beyond the historical selected case.

If the fresh generator effect fails its corrected SCREEN/VALIDATION path, or any apparent benefit is matched by the scope-matched generic control, the state-generation-method paper stops without E3, another backbone, or another benchmark as rescue. If the generator effect passes but First-Fail source/interaction fails, the paper becomes the cleaner matched-evidence state-generation-method paper and explicitly drops failure-evidence superiority.

7 Discussion, Limitations, and Conclusion

The scientific shift is from evidence selection to state regeneration. The initial E2-R17 hypothesis emphasized which rejected trajectories should reach persistent learning. The closed large-sample study and selector intervention do not support a universal witness rule. The more informative observation is downstream: one persistent state is useful when frozen, yet the same updater does not reliably recreate that historical advantage from identical evidence. This changes the primary scientific question from "which failure should we show?" to "what happens when the same evidence is materialized into persistent state again?"

The completed result is instability, not a variance decomposition. The exact-evidence replay contains only two fresh updater realizations in one outcome-selected development stream, and each fresh state was originally observed through a fresh downstream actor execution. Frozen-state remeasurement of the historical strong state partially addresses actor noise, but fresh FF1/FF2 have not yet been remeasured against one common frozen comparator. We therefore avoid claiming a measured generator variance component or that updater variance dominates actor variance. The revised M3 directly targets this residual uncertainty without another updater call.

The compiler is an intervention before it is an algorithmic contribution. Trace-conditioned diagnosis, success/failure contrast, execution-grounded repair, and persistent knowledge accumulation already appear in SkillRevise, SkillCAT, RethinkSkill, and WikiSkill (Liu et al., 2026a; Chen et al., 2026; Liu et al., 2026b; Tang et al., 2026). Three spreadsheet-oriented primitives plus deterministic rendering are not, by themselves, a convincing universal algorithm. Their role in the present paper is to create a fully specified alternative generator under matched evidence. A broader algorithmic claim would require fresh evidence that the controlled generator itself provides a nontrivial advantage beyond generic workflow instructions and state scope.

Generic workflow hygiene and state scope are load-bearing alternatives. Completion, verification, and recovery are sensible spreadsheet-agent instructions independent of any rejected trajectory. Moreover, a trajectory-conditioned compiler may simply emit fewer clauses than a maximal generic checklist. This is why the revised bridge includes both a score-only generic maximum and a trajectory-blind state-scope-matched generic control. If the compiler cannot beat the latter, the trajectory-conditioned diagnosis claim fails even if the compiler beats the native updater.

Failure evidence is no longer a gate on the primary paper. The large-sample evidence already shows that rejected-witness effects are heterogeneous. Accordingly, the revised bridge treats First-Fail-4 superiority and the Evidence$\times$Generator interaction as secondary hypotheses. A matched-evidence generator effect can validate even if First-Fail is no better than Winner. In that case, the cleaner paper is about the reliability of the persistent-state generation interface rather than failure-evidence selection.

The controlled suite is not external validity. The six failure families are generated to permit matched causal interventions and deterministic verification. They should not be interpreted as a representative sample of user spreadsheet workflows. SpreadsheetBench and later workflow-level benchmarks can test transport only after the internal mechanism survives; opening them after a negative fresh generator result would be benchmark shopping.

Selected-case development remains a source of researcher degrees of freedom. The S1 stream and the semantics used to design G1–G3 were selected after historical outcomes. We treat them as a mechanistic sandbox rather than independent confirmation. The revised M3 can reduce actor/comparator ambiguity but cannot erase selected-case bias. The fresh bridge breaks this adaptive loop by freezing tasks, stream assignment, held-out panels, compiler vocabulary, generic controls, and decision rules before bridge outcomes.

Practical implication. Persistent text state is attractive because it is inspectable, but inspectability does not imply reproducibility. Self-evolution studies should therefore report the identity of realized persistent artifacts, not only the trajectory source and final task score. Content-addressed state IDs, same-evidence regeneration checks, and frozen-state actor remeasurement make it possible to distinguish an evidence-policy question from a state-materialization question.

Conclusion. Our current evidence rejects a simple universal story that failed trajectories are better training signals and reveals a narrower phenomenon: in one controlled development case, a historically useful persistent skill remains behaviorally active when frozen, while reconstructed byte-identical evidence does not reliably regenerate the same advantage through the native updater. This is state-regeneration instability consistent with a state-generation bottleneck, not yet a population variance decomposition. We therefore make the state generator an explicit experimental variable, retarget the next frozen-state audit to existing same-evidence states, and prospectively compare native free-form generation with a typed canonical intervention under matched evidence and strong generic controls. The paper succeeds only if that fresh generator question survives its predefined controls; it is not rescued by a new failure selector, a larger benchmark, or another backbone.

Reproducibility Statement

All load-bearing controlled experiments bind task IDs, initial and learned skill SHA-256 values, trajectory evidence hashes, model identity, provider budgets, retry rules, and completion leases. Completed units are not replayed to repair scientific outcomes. The bridge suite is generated deterministically from content-addressed salts, with disjoint SCREEN and VALIDATION panels. The revised manuscript separates the matched-evidence generator gate from failure-evidence and interaction gates, and distinguishes completed scientific evidence from zero-provider design objects.

AI Use Statement

Generative AI tools were used to assist with hypothesis refinement, research software, literature search, adversarial methodology review, manuscript drafting, and artifact inspection. Quantitative claims are checked against persisted experiment receipts and content-addressed artifacts. The authors are responsible for the final scientific claims, citations, and submission.

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A Claim Boundary

Table 2: Current manuscript claim ledger after independent manuscript review.

Status	Statement
Supported	The 48-pair rejected-witness effect is small and heterogeneous under the frozen DeepSeek V2 design.
Supported	On the selected development stream, the historical strong First-Fail persistent state remains directionally useful across two frozen-state actor remeasurements.
Supported	Two fresh free-form updater states generated from reconstructed byte-identical First-Fail evidence do not reproduce the historical First-Fail advantage in their original measurements.
Supported, local	The completed observations show a state-regeneration anomaly consistent with a persistent-state generation bottleneck and rule out evidence disappearance as a sufficient explanation.
Not supported	Rejected failures are universally superior learning evidence.
Not supported	Progress proximity or diagnosticity defines a validated witness-selection law.
Not supported	Generator variance has been isolated from actor variance or estimated at population scale.
Not supported yet	Typed compilation improves downstream utility or beats trajectory-blind state-scope-matched generic construction.
Not supported yet	The regeneration phenomenon generalizes to fresh streams, untouched E3, a second backbone, or a public benchmark.

B Frozen Evidence Summary

DeepSeek V2 Repair2. The frozen design contains 48 pairs, 96 learned states, and 1,728 held-out units. Mean MRW−WIN-C is 0.023148, median 0.013889, standard deviation 0.077970, one-sided sign-flip $p = 0.171875$, and bootstrap interval $[-0.018519, 0.065972]$. The adjudicated verdict is `HOLD_MRW_UNDERPOWERED_OR_HETEROGENEOUS`.

Single-case witness selector. On `e1-tsr-00`, WIN-C/First-Fail/Progress-Fail/Progress-Contrast are 13/18, 17/18, 14/18, and 14/18. The frozen verdict is `S1_SIGNAL_FAIL_STOP_NO_S2`.

Historical frozen-state stability. Two complete remeasurements give First-Fail versus WIN-C of 15/18 versus 14/18 and 16/18 versus 12/18. Verdict: `FIRST_FAIL_FROZEN_STATE_STABILITY_PASS`.

Exact-evidence updater replay. All reconstructed learner-visible packet hashes match the historical First-Fail evidence. Fresh updater realization 1 gives 15/18 versus a fresh WIN-C run of 15/18; realization 2 gives 11/18 versus 12/18. The historical First-Fail state SHA is `97e28b4862...185252`, the two fresh First-Fail state SHAs are `596bd30b49...fdf04f` and `fb5454a27f...36615e`, and all three available WIN-C updater artifacts collapse to the same persistent-state SHA `6df40f6170...c00649`. This byte identity enables the revised M3 common-comparator audit without any new updater state.

C Prospective Stage Status

M2. The deterministic G0–G3 state-semantic intervention is frozen under the existing exactly-once Recovery V3 object. Its incomplete scientific outcome remains sealed at this manuscript checkpoint. M2 tests manual semantic sufficiency only.

M3 review repair. The previous plan to spend M3 on two more remeasurements of G0 plus the M2-selected manual arm is superseded prospectively before any M3 provider execution. The revised M3 freezes the historical First-Fail state, fresh First-Fail states 1 and 2, and the byte-identical common WIN-C state, then contemporaneously remeasures those existing artifacts on the same 18-task panel. No updater call is authorized by the manuscript. Its purpose is to determine whether the FF1–FF2 behavioral separation persists when downstream actor execution and comparator state are controlled more directly.

M4 review repair. The fresh Evidence×Generator bridge remains zero-provider design only. The primary generator hypothesis is now hierarchically separated from First-Fail source superiority and the Evidence×Generator interaction. A positive generator effect can enter VALIDATION without S_C or I passing. Trajectory-conditioned diagnosis additionally requires beating both score-only and scope-matched trajectory-blind generic controls. On VALIDATION only, exactly one additional same-evidence FF4 free-form synthesis per stream is reserved for a direct regeneration diagnostic; it cannot replace the first state.

M5. Untouched E3 confirmation, a second backbone, and public benchmarks have no authority. Failure of the corrected fresh generator path or failure to beat the scope-matched generic control stops the paper without those rescue expansions.

D Bridge Suite Construction

The fresh bridge suite uses new controlled blocks 7–9 and has zero task-ID overlap with the earlier 378-task controlled suite. Blocks 7 and 8 provide update candidates; block 9 provides held-out candidates. From 162 deterministic candidates, hash-based selection freezes 96 update tasks in 12 eight-task streams, 12 SCREEN held-out tasks, 12 disjoint VALIDATION held-out tasks, 12 update integrity reserves, and 30 held-out integrity reserves. Each SCREEN/VALIDATION stream set contains one stream per failure family; each held-out panel contains two tasks per family. Reserved tasks may replace only a pre-execution file-integrity failure, never a model failure or unfavorable scientific outcome.